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MILITARY SPECIFICATION

SHELL, 20MM, HEI, M56A1, METAL PARTS

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the metal parts and metal parts assembly for Shell, 20mm, HEI, M56A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-2550 — General Specification for Ammunition Except Small Arms Ammunition.

MIL-C-490 — Cleaning and Preparation of Ferrous and Zinc Coated Surfaces for Organic Protective Coatings.

MIL-P-10025 — Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification for.

MIL-I-6868 — Inspection Process, Magnetic Particle.

STANDARDS

FEDERAL

Federal Test Method Standard No. 151 — Metals; Test Methods.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

D7553818 — Metal Parts Assembly and Marking Diagram.

FSC 1305

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D7553814 — Cartridge, HEI, 20MM, M56A1.

F7258861 — Carton for Interplant Shipment of Projectiles, 20MM, Assemblies and Metal Parts Assemblies.

PUBLICATIONS

ORDNANCE CORPS

ORD–M608–11 — Procedures and Tables for Continuous Sampling by Attributes.

ORD–M608–14 — Ammunition Ballistic Acceptance Test Methods. Part VIII.

GL 7555816 — Index to Lists of Inspection Gages and Parts Gaged On: Shell, 20MM, HEI, M56A1.

(Copies of specifications, standards, drawing, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with their applicable specifications.

3.2 Parts. The parts and the shell body assembly shall comply with all requirements of the applicable drawings and the specifications shown thereon.

3.3 Ballistic security. No shell body assembly shall show evidence of separation of metal parts in the bore of the test weapon or in flight when tested as specified in 4.5.1.

3.4 Banding. The rotating band blank shall be pressed uniformly into the band seat so as to fill it completely. Prior to application of the rotating band, the band seat shall be cleaned to remove all oil, grease, metal chips, lint, and other foreign matter. Shot or sand blasting is not permitted for cleaning the band seat after knurling.

3.5 Corrosion protection.

3.5.1 *Coating.* The prescribed paint and enamel or lacquer shall be applied in such a manner as to produce a coating of uniform thickness over the entire specified areas. Adhesion and durability shall be as requird by Specification MIL–C–490. The exterior coating shall be dry to the touch prior to packaging for shipment.

3.5.2 *Corrosion resistance.* The coated samples shall show no evidence of rust after being subjected to the test specified in 4.5.5.

3.6 Preproduction samples. Preproduction samples in the quantities specified in 4.3 shall be required prior to the start of regular production.

3.7 Workmanship. The shell body assembly and its parts shall be free from nicks, burrs, sharp edges, cuts, scratches, cracks, seams and porosity and other mutilation, incomplete coating and incomplete, incorrect, or illegible marking, and shall be clean and free of foreign matter. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are

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deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The provisions of Specification MIL–G–2550 apply to this specification. Reference will be made to MIL–STD–109 for definition of inspection terms used herein.

4.3 **Preproduction samples**. Prior to the start of regular production, the contratcor shall submit preproduction samples manufactured by the same methods and equipment which will be utilized in performing the work under contract. The contracting officer will furnish instructions for shipment of the samples to a designated Government laboratory and a proving ground for examination and test. The preproduction samples shall consist of the following:

(a) For ballistic tests at a Government proving ground to ascertain conformance with the acceptance criteria shown in 4.4.3.1.

(1) 4000 shell body assemblies, painted

(b) For physical and dimensional inspection and test at a Government laboratory to ascertain conformance with all other requirements of the applicable drawings and of this specification.

(1) 100 shell body assemblies, painted

(2) 100 shell body assemblies, surfaces not prepared for painting

(3) 100 banded shell bodies, less base cover

(4) 100 unbanded shell bodies

(5) 25 rotating band blanks

(6) 100 base covers

4.3.1 Approval or disapproval of the preproduction samples will be given by the responsible procuring activity. Instructions for the submission of new sample(s) in case of disapproval of the original sample(s) will be issued by the procuring activity. Regular production shall not proceed until the preproduction samples have been approved.

4.4 **Sampling.**

4.4.1 *Lot Formation.* A lot shall consist of shell body assemblies or parts manufactured in accordance with the same drawings and drawing revisions, the same specification and specification revision. A shell body lot shall be restricted to one heat of steel.

4.4.2 *Examination.*

4.4.2.1 Each unit of the lot shall be examined for the critical defects listed in 4.4.4. Any unit found to be defective shall be removed from the lot.

4.4.2.2 Lot sampling and acceptance criteria will be in accordance with MIL–STD–105 or ORD–M608–11, as appropriate. Judgment for lot acceptance will be based on either the individual defect characteristic or on classes of defect characteristics with individual defect characteristic limitations, as approved by the procuring activity prior to the initiation of production, with the following AQL's assigned (see 6.2).

4.4.2.2.1 *Individual defect characteristic basis.*

Major defects	0.25%
Minor defects	0.40%

4.4.2.2.2 *Defect characteristic class basis.*

Major defects	1.50%
Minor defects	2.50%

Individual defect characteristic AQL's associated with the class basis, will be —

Major defects	0.40%
Minor defects	0.65%

4.4.3 *Test inspection.* The tests indicated below will be performed in accordance with the methods specified in 4.5. The number of

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test samples for each lot are as shown in table I. Acceptance criteria are as specified below.

TABLE I.

Test	*Number of Samples*
Ballistic security	250 shell body assemblies, painted.
Soundness	All shell body assemblies, unpainted.
Band tightness	300 banded shell bodies.
Coating weight	Test panel or slug, 4 hours maximum process intervals.
Corrosion resistance	25 shell body assemblies, painted.
Paint adherence	10 shell body assemblies, painted.
Rotating band seat knurling	10 unbanded shell bodies.

4.4.3.1 *Ballistic security.* The sample shall be forwarded to a proving ground designated by the procuring activity. The lot will be rejected, not subject to resist, if any one of the following types of metal parts separations occurs (as evidenced by screen indication or recovered fragments:

(a) Body breakup.
(b) Complete band separation.

A complete band separation is defined as the entire band or a part of the band having band seat serration imprint becoming detached as evidenced by the recovered piece or holes in the screen. The lot will not be penalized for band slivering. Band fragments showing no evidence of the imprint of band seat serrations and having a length no greater than 0.500 inch will be treated as follows:

(a) The lot will be rejected, not subject to retest, if three or more shell body assemblies throw fragments.
(b) The lot will be accepted if two or less shell body assemblies throw fragments.

In the case of fragments greater than the size specified above, the disposition of the lot will be determined by the procuring activity.

4.4.3.2 *Soundness.* The unpainted shell body assemblies found to be defective, when examined as prescribed in 4.5.2, shall be removed from the lot.

4.4.3.3 *Band tightness.* The sample shell body assemblies will be selected from each lot and tested as specified in 4.5.3. If none of the shell body assemblies show evidence of a band gap greater than .004 inch, the lot is acceptable. If any sample shell body assembly fails to meet the requirements for maximum band gap, i.e., .004 inch, the lot will be rejected except that in the event that not more than one shell body fails to meet the requirement, a retest of 600 shell body assemblies is authorized. The lot will be finally rejected if in the combined test and retest a total of two or more shell body assemblies fail to meet the band tightness requirements.

4.4.3.4 *Coating weight.* Failure of the panel or slug to meet the acceptance criteria of MIL-C-490 will cause rejection of the production represented.

4.4.3.5 *Corrosion resistance.* The sample shell body assemblies will be selected from each lot and tested as specified in 4.5.5. If none of the parts being tested fail to meet the requirements for corrosion resistance the parts are acceptable. If any part of the sample tested fails to meet the requirements for corrosion resistance, the lot will be rejected except that in the event of one or two failures to meet the requirements a retest of fifty parts is authorized. The parts will be finally rejected if in the combined test and retest a total of three or more parts fail to meet the corrosion resistance requirements.

4.4.3.6 *Paint Adherence.* The sample shell body assemblies will be selected from each lot and tested as specified in 4.5.6. If none of the assemblies fail to meet the requirement, i.e., if the tape test does not reveal flaking as to expose the phosphate coating on the shell body, the lot is acceptable. If any part of the

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sample tested fails to meet the requirement for paint adherence the lot will be rejected except that in the event of one or two failures to meet the requirement a retest of 20 additional shell body assemblies is authorized. The lot will be finally rejected if in the combined test and retest a total of three or more shell body assemblies fail to meet the paint adherence requirements.

4.4.3.7 *Rotating band seat knurling.* The lot will be rejected, not subject to retest, if one or more sample shell bodies fail to meet the requirement for depth or width of knurling.

4.4.4 *Classification of defects.*

4.4.4.1 Body, prior to crimping base cover *(see Dwg. D7553819 covering a detail of Dwg. D7553818).*

Categories and defects	Method of inspection
Major	
101. Diameter of base cover recess, max	Gage
102. Height of crimping lip from inside of base cover recess, min	Gage
103. Knurling missing in rotating band seat	Visual
104. Diameter of base cover seat	Gage
Minor	
201. Diameter, rotating band seat	Gage
202. Width, rotating band seat	Gage
203. Location, rotating band seat	Gage
204. Length to base cover seat	Gage

4.4.4.2 *Base Cover (see Dwg. B7553820 covering a detail of Dwg. D7553818).*

Categories and defects	Method of inspection
Major	
101. Outside diameter, min	Gage
102. Diameter raised section	Gage
103. Diameter of recess	Gage

4.4.4.3 *Body, rotating band and base cover assembly prior to painting (see Dwg. D7553818 and Dwg. D7553819 covering a detail of Dwg. D7553818).*

Categories and defects	Method of inspection
Critical	
1. Metal defective	Magnetic particle (see 4.4.5.1)
Major	
101. Rotating band damaged	Visual (see 4.4.5.2)
102. Threads not full	Visual
103. Total length, max	Gage
104. Diameter bourrelet, min	Gage
105. Diameter, body rear of rotating band, max	Gage
106. Diameter, rotating band	Gage
107. Pitch diameter of threads	Gage
108. Minor diameter of threads	Gage
109. Length of perfect thread, min	Gage
110. Threads not perpendicular to face	Gage
111. Concentricity, rotating band with bourrelet	Gage
112. Concentricity, cavities with bourrelet	Gage
113. Concentricity, threads with bourrelet	Gage
114. Width of rotating band	Gage
115. Location of crimping groove	Gage
116. Depth of crimping groove	Gage
117. Diameter of small cavity, max	Gage
118. Diameter of large cavity, max	Gage
119. Depth of small cavity, max	Gage
120. Length from front end to rear of rotating band, max	Gage
121. Crimp not flush or below flush (base plate)	Gage
122. Base cover loose	Manual
123. Crimp not full 360 degrees	Visual
124. Corner break, base end of crimping groove	Gage
Minor	
201. Foreign matter	Visual (see 4.4.5.3)
202. Finish improper	Visual
203. Radii missing	Visual
204. Outside Diameter, nose end	Gage

4.4.4.4 *Body assembly, after painting (see Dwg. D7553818).*

Categories and defects	Method of inspection
Major	
101. Paint on rotating band	Visual (see 4.4.5.4)
102. Paint on threads	Visual
103. Diameter of bourrelet, max	Gage
104. Diameter of body at rear of rotating band, max	Gage
105. Base cover missing	Visual

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Minor

201.	Paint not dry (internal paint may remain tacky)	Visual
202.	Interior or exterior paint inadequate	Visual (see 4.4.5.5)
203.	Marking incorrect, incomplete, illegible, or missing	Visual
204.	Weight	Balance
205.	Pool of paint in bottom of cavity	Visual

4.4.4.5 *Packed carton (see Dwg. F7258861).*

	Categories and defects	*Method of inspection*
Major		
101.	Carton damaged	Visual (see 4.4.5.6)
102.	Sealing tape missing or inadequate	Visual
Minor		
	None assigned.	

4.4.5 *Defect standards.*

4.4.5.1 The sample will be examined using the method of 4.5.2. Any shell body of the sample which shows a magnetic particle indication will be sectioned. A sectioned shell body showing a flaw of any kind greater than .005 inch at its greatest dimension will be classed defective.

4.4.5.2 The assembly will be classed defective if the rotating band contains any imperfection, including embedded foreign matter, greater than 0.010 inch in depth and 0.10 inch in length.

4.4.5.3 Any part showing pressure of dirt, oil, grease, corrosion, or other foreign substance will be classed defective.

4.4.5.4 When the protective coating is observed on a bearing surface of the rotating band, the assembly will be classed defective. However, a film of protective coating no thicker than that applied to the shell body appearing on the front and rear edges of the band, will not be considered a defect.

4.4.5.5 Any assembly with a bare spot will be classed defective. However, scratches not penetrating to the base metal will be ignored.

4.4.5.6 "Carton damaged" is defined as any tear, perforation. water damage or other defect which would expose contents or permit or indicate possible damage to the contents.

4.4.6 *Inspection equipment.*

4.4.6.1 Government verification inspection will be accomplished with the use of gages and equipment specified in the "Index to Lists of Inspection Gages and Parts on: Shell, 20MM, HEI, M56A1," GL7555816.

4.5 Test methods.

4.5.1 *Ballistic security.* The shell body assemblies will be assembled into Cartridges, 20MM, HEI, M56A1, in accordance with Drawing D7553814, except that appropriate propellant powder will be used to give a mean pressure of 65,000 ± 2,500 pounds per square inch and will be fired in bursts of 50 rounds in an M39 gun. The method of test will be as prescribed in Army Ordnance Publication ORD–M68–14, Ammunition Ballistic Acceptance Test Methods, Part VIII, Chapter I, Section 4, utilizing electronic fragment detection screens at the 35 and 100 feet range positions. At the beginning of the test, the gun will be at ambient temperature and will be cooled to ambient temperature between bursts.

4.5.2 *Soundness.* The shell assemblies will be examined for soundness using magnetic particle test equipment utilizing the continuous method (wet suspension) and a magnetizing current of 1,000 amperes DC applied to the long axis of the shell assembly. The examination will be performed prior to preparation for painting. The requirements of specification MIL–I–6868 will apply.

4.5.3 *Band tightness.* The test for band tightness will be conducted using a band tightness tester, Dwg. B7259419, referenced

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in GL7555816. The machine will have an initial ram pressure of 200 pounds and a final ram pressure of 3000 pounds.

4.5.4 *Coating weight.* The test for coating weight will be as prescribed in specification MIL-C-490. Test panels or slugs of the same material as the shell body will be used for the determination of the weight of the coating. When used, slugs will be of such configuration that their entire area may be readily measured. The test specimens will be treated at the same time and in the same manner as the shell bodies.

4.5.5 *Corrosion resistance.* The test of corrosion resistance will be conducted in accordance with Method 811 of Federal Test Methods Standard No. 151 — Metals, Test Methods. The base of the assemblies up to and including the rotating band will be masked out by a wax coating or other suitable material. Exposure time to salt spray will be 24 hours.

4.5.6 *Paint adherence.* The test for paint adherence will be conducted in accordance with Specification MIL-C-490.

4.5.7 *Rotating band seat knurling.* Measurements will be made using the gage and measuring device listed in GL7555816 at three locations approximately 120° apart on each sample body (see 6.3).

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level C.* Painted shell body assemblies shall be packed in accordance with Drawing F7258861.

5.2 Marking.

5.2.1 *Level C.* Marking on packs of shell body assemblies shall be in accordance with Specification MIL-P-10025.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Place of inspection, if not place of manufacture.

6.2 **AQL's.** The optional use of AQL values for either individual defects or classes of defects, as approved by the procuring activity prior to the initiation of production, is authorized to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the process.

6.3 **Rotating band seat knurling.** Visual judgment may be substituted for measurement, using the latter for borderline cases only. The minimum width of rotating band seat is greater than the minimum width of knurling.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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